Yueyang Pan Ph.D. Candidate

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Education

EDIC Computer and Communication Sciences, EPFL Lausanne, Switzerland
Ph.D in computer science 2022 - 2027 *(expected)*

- Advisor: Prof. Sanidhya Kashyap
- Research area: Operating Systems, LLM inference and training Systems

EECS, Peking University Beijing, China
B.S. in Computer Science and Technology; Minor in Finance 2018 - 2022

- GPA: 3.83/4.00, Rank: 12/230.
- Graduate with *Summa cum laude*
- Turing Class Honor Program
- Advisor: Prof. Chenren Xu
- Research area: Computer Networks and 5G Wireless Communication

Research Interests

My research focuses on state (page, KV cache, agentic memory) management across the memory hierarchy, from OS kernels, LLM serving frameworks to agent memory systems. The trade-off between fetching and regenerating state recurs at every layer, and my work aims to make memory-bound systems scale by resolving this mechanism and policy design question coherently across the stack.

Publications

1. **CacheWeave: Learning to Recompute for Efficient Position-Independent KV Caching**
   Yueyang Pan, Qianchao Zhu, Yuliang Liu, Teng Ma, Sanidhya Kashyap
   https://arxiv.org/abs/2611.01735

2. **NanoMem: Iterative Evidence Synthesis for Temporal-Causal Reasoning in Agent Memory**
   Yueyang Pan, Youliang Zhu, Nikhil Sarda, Haodong Zheng, Antoine Bosselut, Sanidhya Kashyap
   Under submission to ICLR 2027

3. **Scalable Far Memory: Balancing Faults and Evictions**
   Yueyang Pan*, Yash Lala*, Musa Unal, Yujie Ren, Seung-seob Lee, Yizhou Shan, Abhishek Bhattacharjee, Anurag Khandelwal, Sanidhya Kashyap (*Equal Contribution)
   ACM Symposium on Operating Systems Principles (SOSP 2025)

4. **Tolerate It if You Cannot Reduce It: Handling Latency in Tiered Memory**
   Musa Unal, Vishal Gupta, **Yueyang Pan**, Yujie Ren, Sanidhya Kashyap
   ACM SIGOPS 20th Workshop on Hot Topics in Operating Systems (HotOS XX)

5. **Transparent Multicore Scaling of Single-Threaded Network Functions**
   Lei Yan, **Yueyang Pan**, Diyu Zhou, George Candea, Sanidhya Kashyap
   European Conference on Computer Systems (EuroSys 2024)

6. **Monarch: A Fuzzing Framework for Distributed File Systems**
   Tao Lyu, Liyi Zhang, Zhiyao Feng, **Yueyang Pan**, Yujie Ren, Meng Xu, Mathias Payer,

Sanidhya Kashyap
USENIX Annual Technical Conference (ATC 2024)

7. **Ship your Critical Section, Not Your Data: Enabling Transparent Delegation with TCLOCKS**
Vishal Gupta, Kumar Kartikeya Dwivedi, Yugesh Kothari, **Yueyang Pan**, Diyu Zhou, Sanidhya Kashyap
USENIX Symposium on Operating Systems Design and Implementation (OSDI 2023)

8. **The First 5G-LTE Comparative Study in Extreme Mobility**
Yueyang Pan*, Ruihan Li*, Chenren Xu (*Equal Contribution)
ACM on Measurement and Analysis of Computing Systems (SIGMETRICS 2022)

9. **Critique of "A parallel framework for constraint-based Bayesian network learning via Markov blanket discovery"**
Jiaqi Si, Junyi Guo, Zhewen Hao, Wenyang He, Ruihan Li, **Yueyang Pan**, Zhenxin Fu, Chun Fan
IEEE Transactions on Parallel and Distributed Systems (TPDS 2022)

10. **Critique of "MemXCT: Memory-Centric X-Ray CT Reconstruction With Massive Parallelization"**
Zejia Fan, Yuchen Gu, Zhewen Hao, **Yueyang Pan**, Pengcheng Xu, Yuxuan Yan, Fangyuan Yang, Zhenxin Fu, Yun Liang
IEEE Transactions on Parallel and Distributed Systems (TPDS 2021)

EXPERIENCES

Rednote | Hangzhou 2026.07 -
Mentor: Qianchao Zhu
- Ace Talent Program
- CacheWeave: Training-aware position independent KV cache mechanism for note and agentic search
- DemystifyPIC: understand training-aware position independent KV cache at scale
- Efficiency of the pretraining framework

Meta | London 2025.06 - 2025.09
Mentor: Nikolay Beloborodov, Usama Arif and SJ Park
- Enabling memory allocation profiling across meta fleet ($\sim 10m$ machines)
- Feature integration into Strobelight profiler and OOM workflow
- Debugging new kernel deployment and new hardware rollout with the feature
- Multiple patches upstream (DAMON and memory allocation profiling)

Alibaba | Hangzhou, China 2022.05 - 2022.09
Mentor: Nie Hao and Junchen Guo
- Develop testing framework for 5G Cloud Solutions
- Test and analyze the performance of different solutions

PKU Super Computing Team | Beijing, China 2021.04 - 2022.06
Mentor: Zhenxin Fu, Yun Liang and Chun Fan
- Participate in ASC Student Supercomputer Challenge (ASC) 20-21, SC Student Cluster Competition (SCC) 20, 21

Projects

Operating Systems

Mage 2023.09 - 2025.07

Mage extends the feasibility of memory offloading. We find existing solutions have severe multi-core scalability issues, preventing many apps from being offloaded. Our system, Mage, adopts design principles, such as asynchronous decoupling, pipelined execution,and contention avoidance, to address the issues and realizes them on both Linux and LibOS

NFOS 2021.10 - 2023.11

NFOS is a combination of programming model, runtime, and profiler for developing scalable network functions (NFs) on multicore servers. NFOS insulates developers from writing concurrent NF code as it is bug-prone and hard to scale. We show that serial, stateful NFs run atop NFOS achieve scalability on par with hand-optimized counterparts in Cisco VPP.

TCLock 2021.10 - 2022.12

TCLock is a family of locking protocols using transparent delegation, where a lock waiter automatically encodes its critical section information on its stack and notifies the combiner (lock holder). The combiner executes the shipped critical section on the waiter's behalf using a lightweight context switch. TClock requires zero app modification with superior performance.

Machine Learning Systems

MegaLLM 2024.08 -

MegaLLM is a new decoupled architecture with a memory daemon to manage KV caches across storage tiers (including SSD, DRAM and VRAM), minimizing transmission costs in the cluster.

Side projects

PDFPal 2026.01 -

A lightweight PDF reader with LLM chat panel, markdown note rendering support and offline citation information card across platforms (MacOS, and iPadOS)

Open Source

SGLang 2025.01 -

Work on Multi-tiering KV Cache and MLLM. Bug fix and performance optimizations

Linux 2025.06 -

Work on memory management (DAMON and memory allocation profiling)

vLLM 2024.07 - 2024.12

Work on Multi-tiering KV Cache

UCCL 2026.06 -

Working on supporting slingshot/CXI

Apertus v2 2026.03 -

Pretraining and post training performance

Teaching

Advanced Operating Systems, CS477, EPFL 2025.09

Data-intensive Systems, CS300, EPFL 2025.02

Advanced Operating Systems, CS477, EPFL 2024.09

Data-intensive Systems, CS300, EPFL, 2024.02
Introduction to Operating Systems, CS323, EPFL 2023.09
Introduction to Operating Systems, CS323, EPFL 2022.09
Computer Networks (Honor Track), 04832250, Peking University 2021.09
Summer Reading Group, Turing Class, Peking University 2019.08

Awards and Honors

- **Teaching Assistant Award**, EDIC, EPFL 2024.12
- **Outstanding Graduates of Bejing**, Beijing 2022.06
- **Outstanding Graduates of Peking University**, Peking University 2022.06
- **China National Scholarship**, China 2021.09
- **Merit Student**, Peking Univeristy 2021.09
- **ASC Student Supercomputer Challenge 20-21 4th Place**, Worldwide 2021.05
- **John Hopcroft Scholarship**, Turing Class 2020.09
- **Schlumberger Scholarship**, Peking University 2019.10

Skills

Languages: Chinese, English, French (A2).
Programming: `C++/C`, `Python`, `Shell`, `CUDA`, `Linux`.